WQF7004
Data Analytics in Artificial Intelligence

Understanding your dataset and asking questions

- don’t know exactly what they want.
- might have a vague idea of what the problem is and what they need as a solution.

Business people

Data scientists

What Questions to Ask?

Need to understand the problem & translate business needs into technical model requirements.

- ✔ What business problem are you trying to solve?
- ✔ What’s the pain points in your current situation?
- ✔ What’s your use case? Can you give me an example?
- ✔ What’s the end product you wish to deliver?
- ✔ If we want to start small, are there any specific use cases you want to take first?

Good Question

Let's say you want to find out how rich people are across the world and how that differs in different geographic regions.

#1: Specific Terminology

Are the terms you are asking specific to that domain?

A good first start or "pre-question" would ask "How do economists measure the wealth of individuals?" or "what indicators measure individual wealth?" These might lead you to terms like GDP per capita, GNP per capita, etc.

#2: Specific Scope

You need to have a scope, or you'll have to ask a series of follow up questions. For example, are you interested in particular countries, regions or continents? Are you concerned about their current wealth or you want data from xyz years ago?

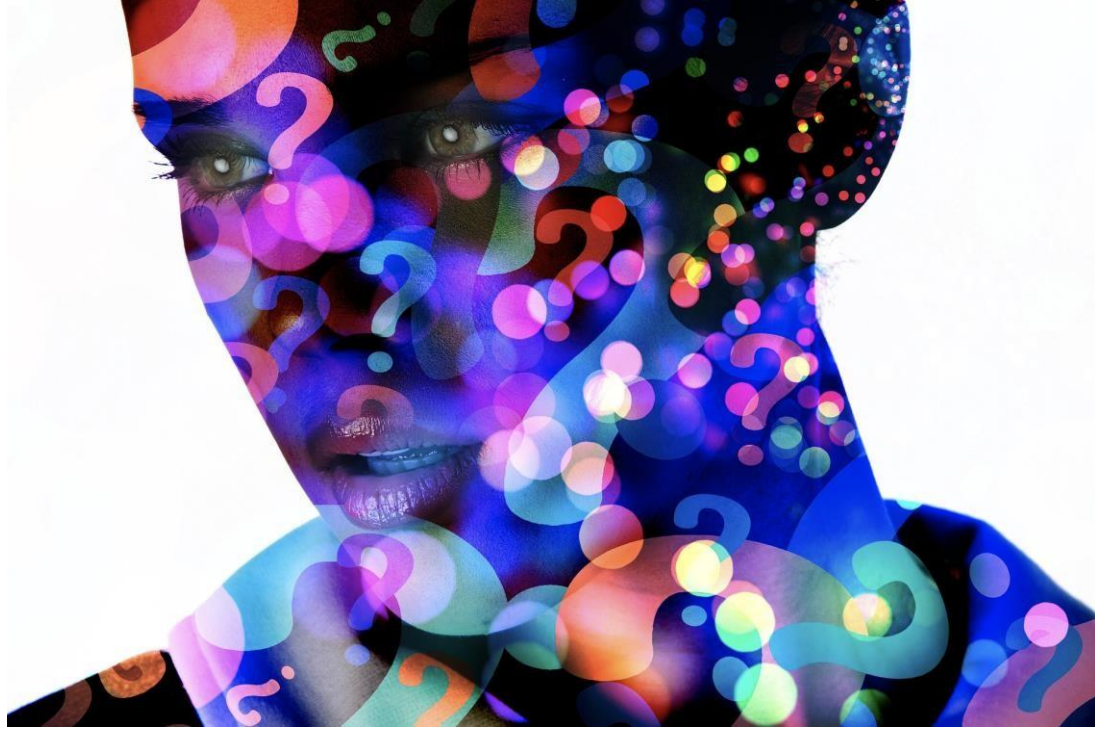
#3: Specific Source

Source refers to a broad sense of whom you're asking this question. Is it reasonable?

https://towardsdatascience.com/how-to-ask-good-questions-be41b517c1b

https://towardsdatascience.com/how-to-ask-questions-data-science-can-solve-e073d6a06236

- Traditional **hypothesis-driven** science was based on asking specific questions of the world and then generating the specific data needed to confirm or deny it.
- Today **data-driven** science focuses on generating data on a previously unheard of scale or resolution, in the belief that new discoveries will come as soon as one is able to look at it.



Both ways of thinking will be important to us:

✔ Given a **problem**, what available **data** will help us answer it?

✔ Given a **data set**, what interesting **problems** can we apply it to?

For a client, a data science project usually begins in one of two ways.
One, the client says,"We have this business situation, and we need help resolving it."
Or two, a client says,"We have this data set, and we think we could use it to benefit our business." Think of it as clients who want to know *an answer* and clients who want to know *what's possible*.

Examples of Questions

- **Hypothesis testing** tells us if our data applies to new situations. *"Do cat pictures drive more traffic than dog pictures?"*
- **Scenario analysis** analyzes many possible future outcomes under various conditions. We create many possible scenarios and then predict what will happen. *"What would happen if we raise the price of our product?"*
- **Optimization** generally asks simple, yet hard to answer, maximization and minimization questions. *"What supply routes minimize the cost of delivering packages?"*
- **Reinforcement learning** observes data and optimizes an outcome in real-time. *"When should I click to survive in the game Flappy Bird?"*